

Supplement: The Exact Temperature of Exponential-Weight Aggregation

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1 Exact Risk Identity

We first prove the algebraic identity used in the lower bound. For any two real predictions f_1, f_2 , any $w \in [0, 1]$, and any label y ,

$$\begin{aligned} & \{wf_1 + (1-w)f_2 - y\}^2 \\ &= w(f_1 - y)^2 + (1-w)(f_2 - y)^2 \\ & \quad - w(1-w)(f_1 - f_2)^2. \end{aligned} \quad (1)$$

This follows by expanding both sides.

Proof of Proposition 2 in the main paper. On the two-point construction, $\mathcal{R}_{P_n}(f_2) - \mathcal{R}_{P_n}(f_1) = ac/\sqrt{n}$ and $(f_1 - f_2)^2 = h^2$ almost surely. Apply (1) with $w = W_n$, take population expectation conditional on the training sample, and subtract $\mathcal{R}_{P_n}(f_1)$. This gives

$$\frac{ac}{\sqrt{n}}(1 - W_n) - h^2 W_n(1 - W_n).$$

The cumulative empirical loss difference is aZ_n , so the definition of the Gibbs weights gives $W_n = \text{sigm}(aZ_n/T)$. $\square$

2 Triangular Binomial Limits

Fix $c > 0$. For every sufficiently large n , let $\epsilon_{n,1}, \dots, \epsilon_{n,n}$ be i.i.d. random signs with

$$P(\epsilon_{n,i} = 1) = \frac{1}{2} + \frac{c}{2\sqrt{n}}, \quad P(\epsilon_{n,i} = -1) = \frac{1}{2} - \frac{c}{2\sqrt{n}},$$

and put $Z_n = \sum_i \epsilon_{n,i}$. Let ϕ and Φ be the standard normal density and distribution functions.

Lemma 1 (Global and local limits). *The following claims hold.*

1. $Z_n/\sqrt{n}$ converges in distribution to $N(c, 1)$.
2. For each fixed integer k along sample sizes satisfying $n \equiv k \pmod{2}$,

$$\sqrt{n} P(Z_n = k) \longrightarrow 2\phi(c).$$

3. There is $C_c < \infty$ such that, for all sufficiently large n and every integer k ,

$$\sqrt{n} P(Z_n = k) \leq C_c.$$

Proof. The row mean is $\mathbb{E}\epsilon_{n,i} = c/\sqrt{n}$ and the row variance is $1 - c^2/n$. The bounded triangular-array central limit theorem proves the first claim.

For the second and third claims, write $B_n = (n + Z_n)/2$. Then B_n is binomial with success probability $p_n = 1/2 + c/(2\sqrt{n})$. If k has the correct parity,

$$P(Z_n = k) = P\left(B_n = \frac{n+k}{2}\right).$$

For fixed k , put $d_{n,k} = (n+k)/2 - np_n$ and $v_n = np_n(1-p_n)$. Stirling's formula for the binomial mass gives

$$\begin{aligned} P\left(B_n = \frac{n+k}{2}\right) &= \frac{1 + o(1)}{\sqrt{2\pi v_n}} \exp\left\{-\frac{d_{n,k}^2}{2v_n}\right\} \\ &= \frac{2\phi(c) + o(1)}{\sqrt{n}}. \end{aligned}$$

For the uniform claim, the maximum binomial point mass is at most $C/\sqrt{np_n(1-p_n)}$. Since $p_n(1-p_n)$ is bounded away from zero for all sufficiently large n , the required bound follows. Incorrect parity has probability zero. $\square$

We next prove the two limits in the main text. Define

$$q(u) = \text{sigm}(u)\{1 - \text{sigm}(u)\}.$$

Notice that $0 \leq q(u) \leq e^{-|u|}$.

Proof of Lemma 3 in the main paper. The triangular central limit theorem gives $U_n := Z_n/\sqrt{n} \Rightarrow U \sim N(c, 1)$. For every fixed $\eta > 0$,

$$\text{sigm}(-a\sqrt{n}U_n/T)$$

converges uniformly to $\mathbf{1}\{U_n < 0\}$ on $|U_n| \geq \eta$. The probability of $|U_n| < \eta$ has limit at most $P(|U| \leq$

η), which vanishes with η . Since the variables are bounded, it follows that

$$\mathbb{E}[1 - W_n] = \mathbb{E} \text{sigm}(-aZ_n/T) \longrightarrow P(U < 0) = \Phi(-c).$$

Now restrict to sample sizes of parity r . Put $q_{j,r} = q(a(2j+r)/T)$. By Lemma 1, for every $j \in \mathbb{Z}$,

$$\sqrt{n} P(Z_n = 2j + r) \longrightarrow 2\phi(c).$$

Moreover,

$$\sqrt{n} P(Z_n = 2j + r) q_{j,r} \leq C_c e^{-a|2j+r|/T},$$

and the right side is summable over j . Dominated convergence for series gives

$$\begin{aligned} \sqrt{n} \mathbb{E}[W_n(1 - W_n)] &= \sum_{j \in \mathbb{Z}} \sqrt{n} P(Z_n = 2j + r) q_{j,r} \\ &\longrightarrow 2\phi(c) \sum_{j \in \mathbb{Z}} q_{j,r}. \end{aligned}$$

This is the claimed limit. $\square$

3 Positive Limits Below Four

We include all details needed to choose the constants uniformly over both sample parities.

Lemma 2 (Shifted logistic Riemann sums). *For fixed $T > 0$, $r \in \{0, 1\}$, and $a > 0$, put*

$$S_r(a, T) = \sum_{j \in \mathbb{Z}} q\left(\frac{a(2j+r)}{T}\right).$$

Then

$$\lim_{a \downarrow 0} \frac{2a}{T} S_r(a, T) = 1.$$

The convergence holds uniformly over $r \in \{0, 1\}$.

Proof. The function q is continuous, nonnegative, and integrable. It is the derivative of sigm , so its integral over the real line is one. The displayed expression is the Riemann sum for q on a grid of spacing $2a/T$. The two values of r correspond to the unshifted grid and its half-grid translate. Both converge to the same integral. Uniformity is immediate because there are only two shifts. $\square$

Lemma 3 (Gaussian choice). *For every $T < 4$, there is $c > 0$ such that*

$$2c\Phi(-c) - \frac{T}{2}\phi(c) > 0.$$

Proof. Integration by parts gives the standard inverse Mills asymptotic

$$\Phi(-c) = \frac{\phi(c)}{c} \{1 + O(c^{-2})\}.$$

Therefore $c\Phi(-c)/\phi(c) \rightarrow 1$. Choose c so that this ratio exceeds $T/4$. $\square$

Proof of the lower half of Theorem 1. Fix $T < 4$ and choose c by Lemma 3. For $h \in (0, 1)$, let $a = h(2 - h)$ and

$$L_r(h) = ac\Phi(-c) - 2h^2\phi(c)S_r(a, T).$$

Lemma 2 and $a/h \rightarrow 2$ imply

$$\begin{aligned} \frac{L_r(h)}{h} &= (2 - h)c\Phi(-c) - 2h\phi(c)S_r(a, T) \\ &\longrightarrow 2c\Phi(-c) - \frac{T}{2}\phi(c) > 0. \end{aligned}$$

Choose one fixed $h > 0$ small enough that $L_0(h) > 0$ and $L_1(h) > 0$. Proposition 2 and Lemma 3 in the main paper show that, along parity r ,

$$\sqrt{n} \mathbb{E}[\text{excess risk}] \longrightarrow L_r(h).$$

With $\kappa = \frac{1}{2} \min\{L_0(h), L_1(h)\}$, the expected excess is at least $\kappa/\sqrt{n}$ for every sufficiently large n . The same c, h, f_1, f_2 work for every such n . Only the two input probabilities change.

For a range bound $b > 0$, multiply Y, f_1, f_2 by b . Every squared loss, risk difference, and excess risk is multiplied by b^2 , while Gibbs weights at temperature T become the normalized weights at temperature T/b^2 . Hence the lower bound holds whenever $T < 4b^2$, with value $\kappa b^2/\sqrt{n}$. The upper half is Theorem 1 of Høgs-gaard et al. (2026). This completes the exact phase theorem. $\square$

4 Prior-Weighted Exact Phase

Proof of Corollary 4 in the main paper. We first prove the upper bound. Let $N = n + 1$ and write

$$S_j = \sum_{i=1}^N (f_j(X_i) - Y_i)^2, \quad p_j = \frac{\pi_j e^{-S_j/T}}{\sum_k \pi_k e^{-S_k/T}}.$$

After observation i is removed, the prior-weighted Gibbs probabilities are

$$p_j^{(-i)} = \frac{p_j e^{(f_j(X_i) - Y_i)^2/T}}{\sum_k p_k e^{(f_k(X_i) - Y_i)^2/T}}.$$

When $T \geq 4b^2$, the squared-loss tilting inequality of Høgs-gaard et al. (2026) gives

$$\sum_{i=1}^N \left\{ \sum_j p_j^{(-i)} f_j(X_i) - Y_i \right\}^2 \leq \sum_j p_j S_j.$$

The Gibbs variational identity and nonnegativity of relative entropy give, for every j ,

$$\begin{aligned} \sum_k p_k S_k &= -T \log \left(\sum_k \pi_k e^{-S_k/T} \right) - \text{TKL}(p \| \pi) \\ &\leq S_j + T \log(1/\pi_j). \end{aligned}$$

Take expectations. By exchangeability, every term on the left has expectation $\mathbb{E}_{D_n} \mathcal{R}_P(\hat{f}_{T,\pi})$. The expected value of S_j is $N \mathcal{R}_P(f_j)$. Dividing by N and minimizing over j proves the stated oracle inequality.

For the converse, it is enough to work at $b = 1$. Fix $\pi_1, \pi_2 > 0$ and put $\lambda = \log(\pi_1/\pi_2)$. The prior-weighted aggregate places weight

$$W_{n,\pi} = \text{sigm}(aZ_n/T + \lambda)$$

on f_1 . Proposition 2 in the main paper is algebraic, so its risk decomposition remains valid with W_n replaced by $W_{n,\pi}$.

Since λ is fixed, the triangular-array central limit argument in the proof of Lemma 3 gives

$$\mathbb{E}[1 - W_{n,\pi}] \rightarrow \Phi(-c).$$

Along sample sizes of parity r , the same local limit and dominated-convergence argument gives

$$\sqrt{n} \mathbb{E}[W_{n,\pi}(1 - W_{n,\pi})] \rightarrow 2\phi(c) S_{r,\lambda}(a, T),$$

where

$$S_{r,\lambda}(a, T) = \sum_{j \in \mathbb{Z}} q \left(\frac{a(2j+r)}{T} + \lambda \right).$$

For every fixed λ , this shifted lattice satisfies

$$\frac{2a}{T} S_{r,\lambda}(a, T) \rightarrow \int_{-\infty}^{\infty} q(u) du = 1 \quad (a \downarrow 0)$$

for both parities. The proof of Theorem 1 therefore applies without change: first choose c with $2c\Phi(-c) > (T/2)\phi(c)$, then choose one fixed $h > 0$ small enough to make both parity limits positive. Scaling by b completes the proof. $\square$

5 Sharpness of the Tilting Inequality

We prove the necessity part of Proposition 5. Sufficiency at $T \geq 4$ is Lemma 2 of Høggsgaard et al. (2026).

Fix $s \in (0, 1)$ and let U have probability $1 - \varepsilon$ at s and probability ε at one. Let p denote this law and define its exponential tilt by

$$q(du) = \frac{e^{u^2/T} p(du)}{\mathbb{E}_p e^{U^2/T}}.$$

Put $A = e^{s^2/T}$ and $B = e^{1/T}$. Direct differentiation at $\varepsilon = 0$ gives

$$\begin{aligned} \mathbb{E}_q U &= \frac{(1 - \varepsilon)sA + \varepsilon B}{(1 - \varepsilon)A + \varepsilon B} \\ &= s + \varepsilon \frac{B}{A} (1 - s) + O(\varepsilon^2) \end{aligned} \quad (2)$$

and

$$\begin{aligned} (\mathbb{E}_p U^2)^{1/2} &= \{(1 - \varepsilon)s^2 + \varepsilon\}^{1/2} \\ &= s + \varepsilon \frac{1 - s^2}{2s} + O(\varepsilon^2). \end{aligned} \quad (3)$$

The first-order coefficient in (2) exceeds that in (3) exactly when

$$e^{(1-s^2)/T} > \frac{1+s}{2s},$$

or equivalently

$$T < A(s) := \frac{1 - s^2}{\log\{(1+s)/(2s)\}}. \quad (4)$$

In that case $\mathbb{E}_q U > (\mathbb{E}_p U^2)^{1/2}$ for all sufficiently small positive ε , contradicting the tilting inequality.

Let $s = 1 - d$ and send $d \downarrow 0$. Then

$$1 - s^2 = 2d + O(d^2), \quad \log \frac{1+s}{2s} = \frac{d}{2} + O(d^2),$$

so $A(s) \rightarrow 4$. For every $T < 4$, choose s close enough to one that $T < A(s)$, then choose $\varepsilon > 0$ small enough to obtain a violation. Thus universal validity requires $T \geq 4$.

For predictions in $[0, b]$, replace U by bU . The tilt exponent becomes $b^2 U^2/T$, so the normalized temperature is T/b^2 . The threshold is therefore $4b^2$.

References

Høggsgaard, M. M., Rebeschini, P., and Wegel, T. (2026). Aggregation with exponential weights is optimal in expectation. *arXiv preprint arXiv:2607.02247*.